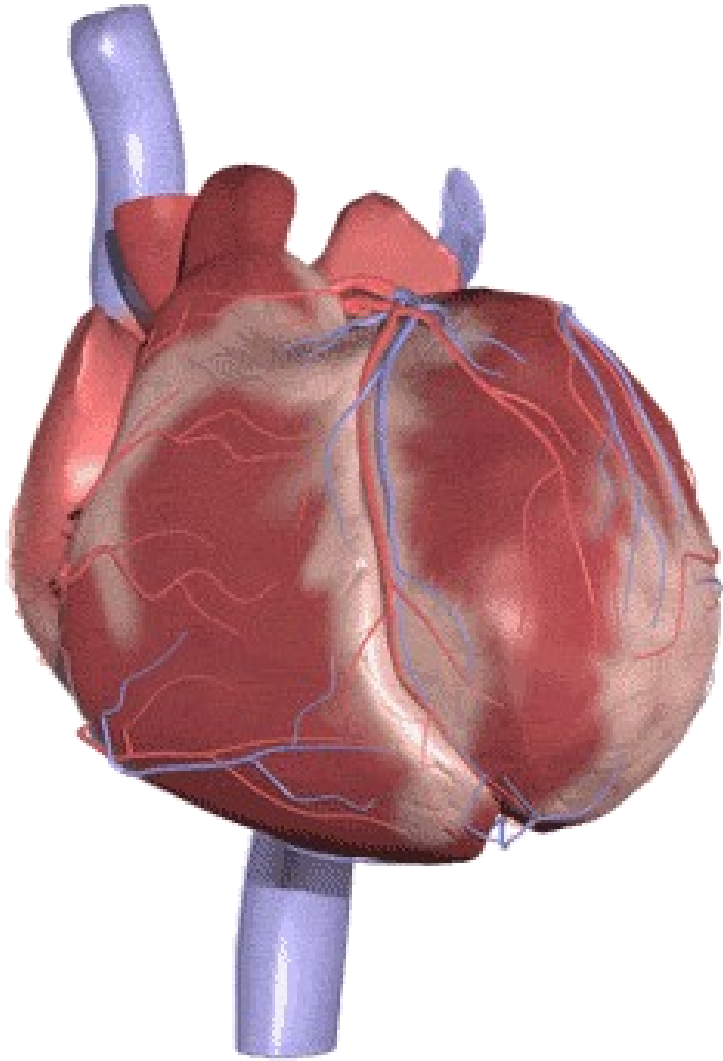
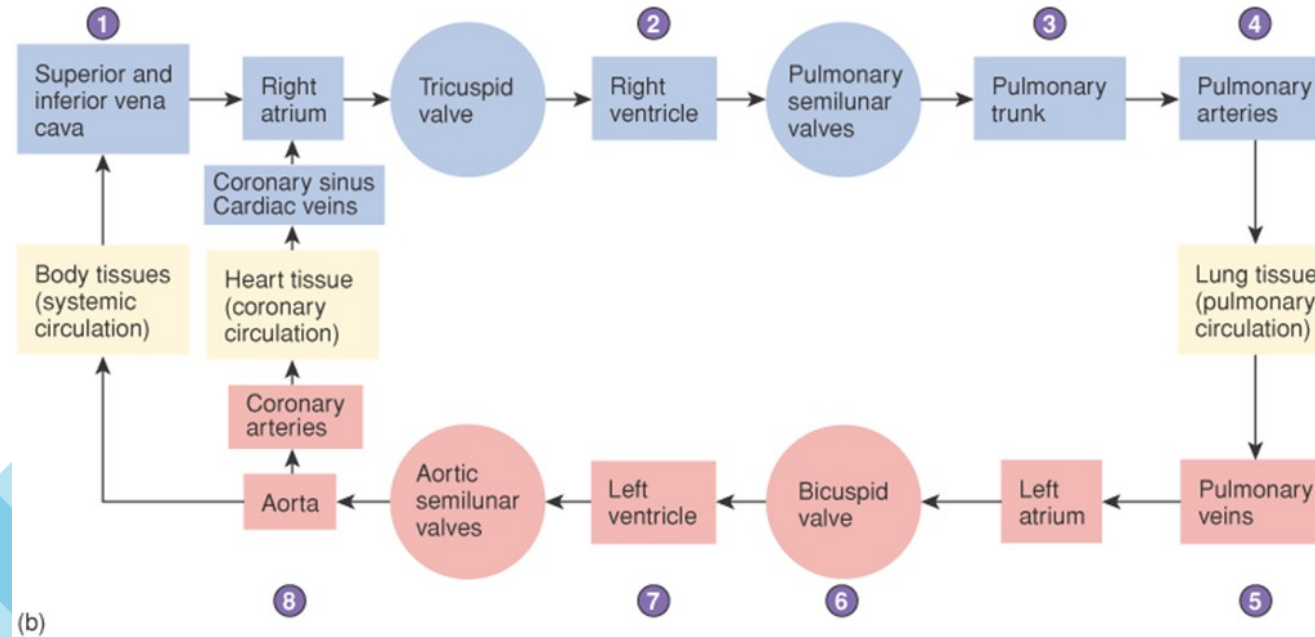
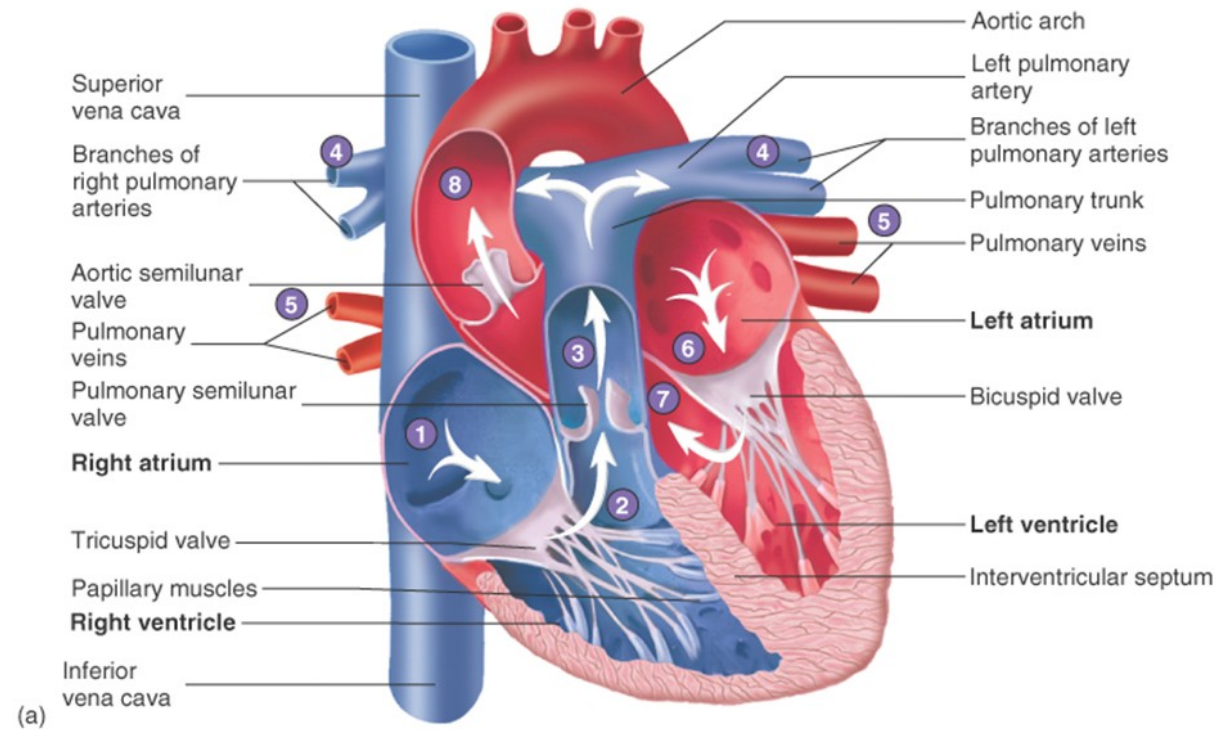


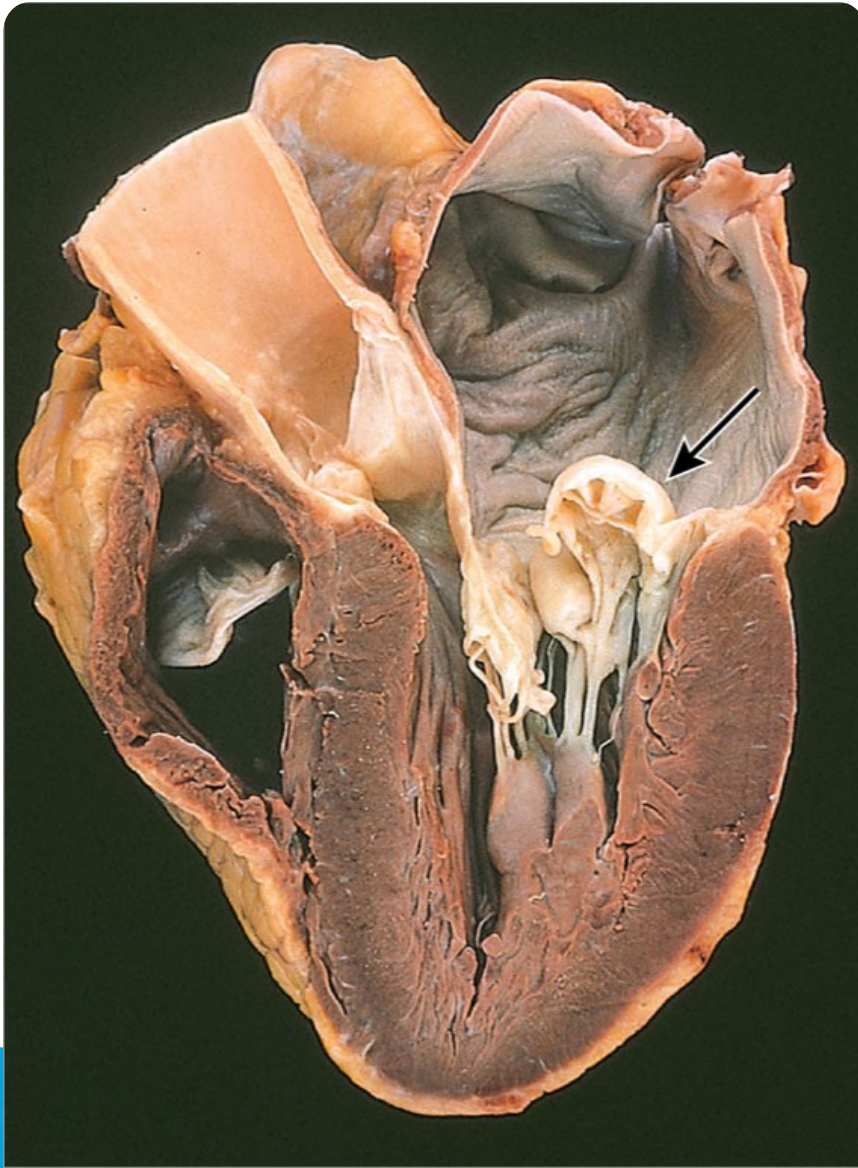


BIO262

Physiology

Module 10 – Cardiovascular System



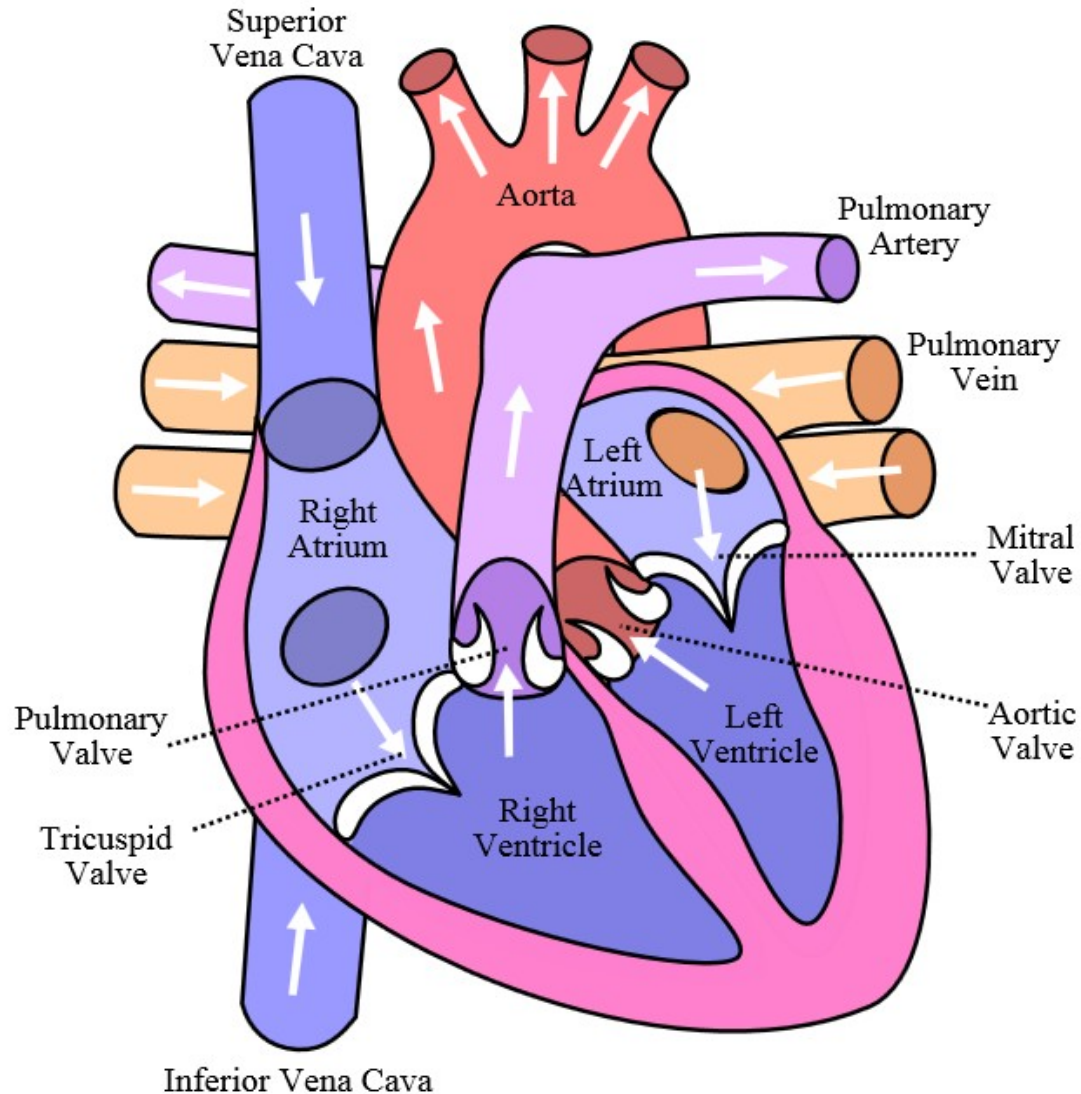


From Kumar V, Abbas AK, Fausto N: Robbins and Cotran pathologic basis of disease, ed 8, Philadelphia, 2010, Saunders.

Blood Flow Through the Heart:

- **Right Atrium:** receives deoxygenated blood from the body
- **Right Ventricle:** receives deoxygenated blood from the right atrium via the tricuspid valve
- **Lungs:** receives deoxygenated blood from the right ventricle via the pulmonic valve and pulmonary artery
- **Left Atrium:** receives oxygenated blood from the lungs via the pulmonary vein
- **Left Ventricle:** receives oxygenated blood from the left atrium via the mitral valve and pumps it to the body via the aortic valve

Valves



- What is the purpose of valves?
 - To prevent backflow, blood should only move in one direction.

Totally Procrastinated My Assignment!

- Tricuspid: prevents backflow into the right atrium during systole
- Pulmonic: prevents backflow into the right ventricle during diastole
- Mitral: prevents backflow into the left atrium during systole
- Aortic: prevents backflow into the left ventricle during diastole

S1 (Lub): AV valves closing (mitral and tricuspid)

S2 (Dub): Semilunar valves closing (pulmonic and aortic)

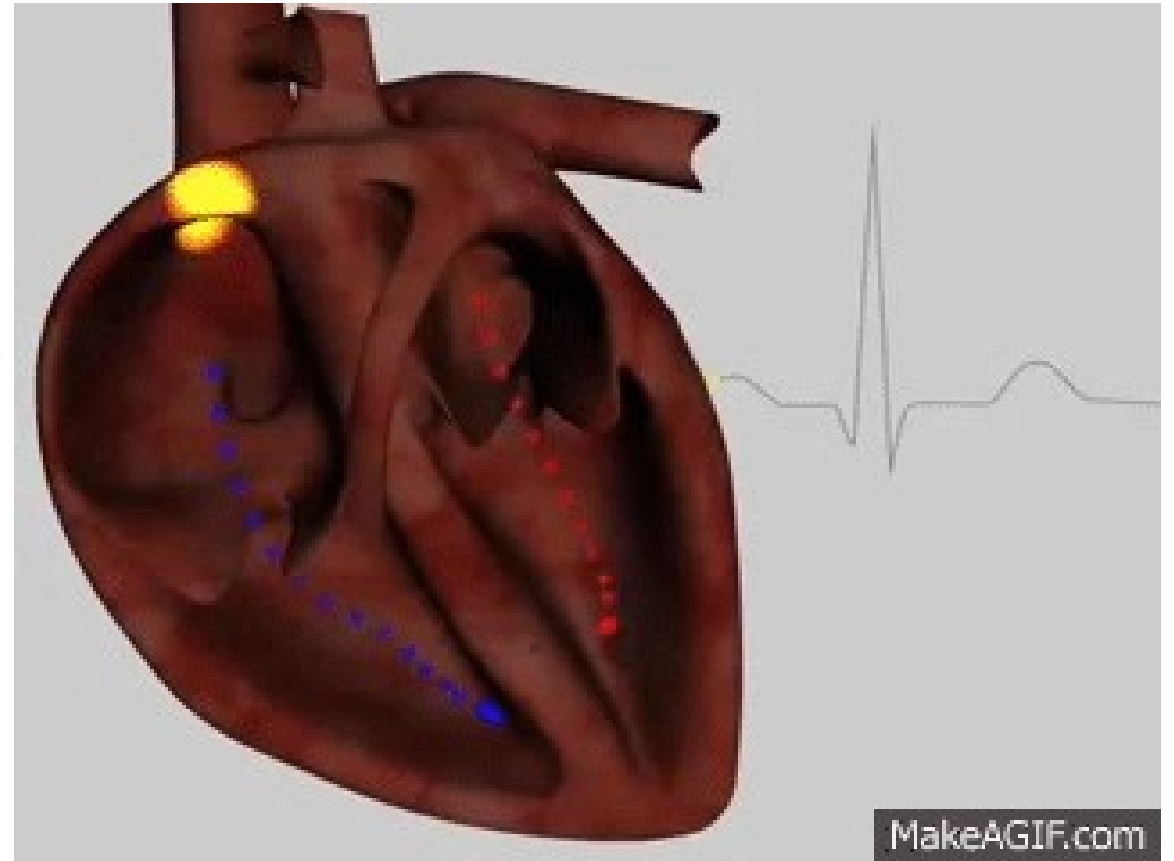
Cardiac Conduction

1. Sinoatrial Node (SA):

- Depolarizes faster than any part of myocardium
- **PACEMAKER** cells that initiate the heartbeat

2. Atrioventricular Node (AV):

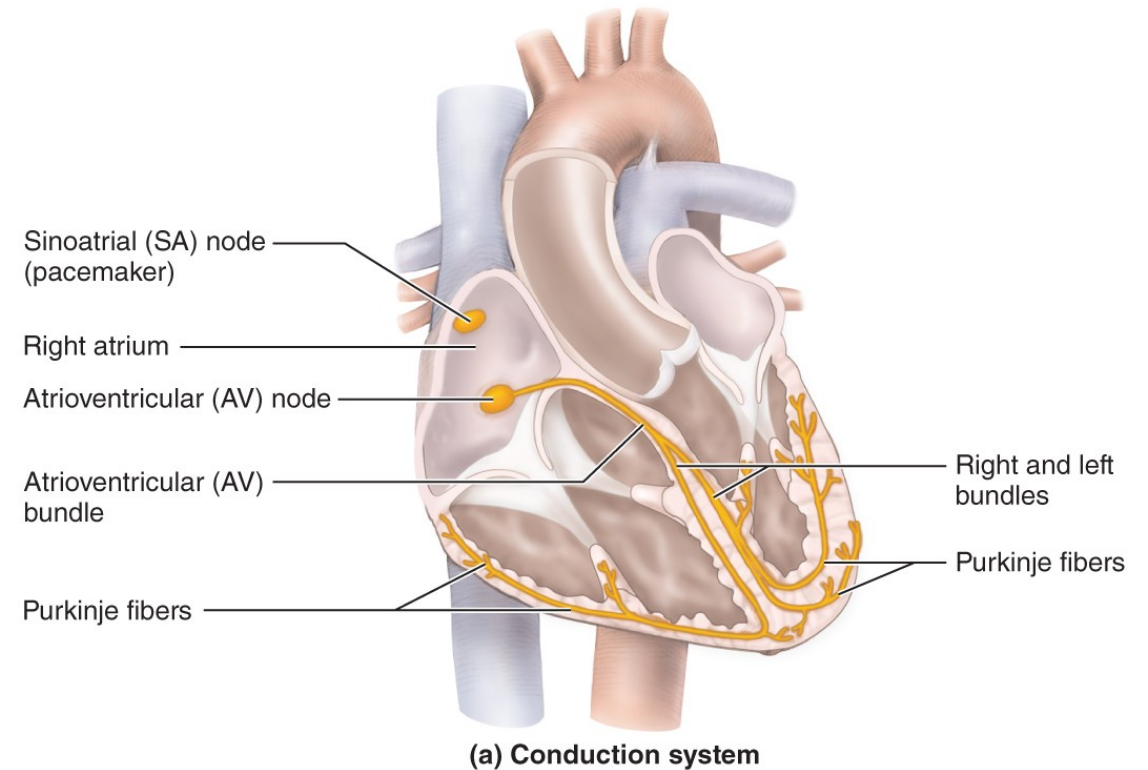
- Delays impulse (slows it) so that the atria can contract before the ventricles



Cardiac Conduction

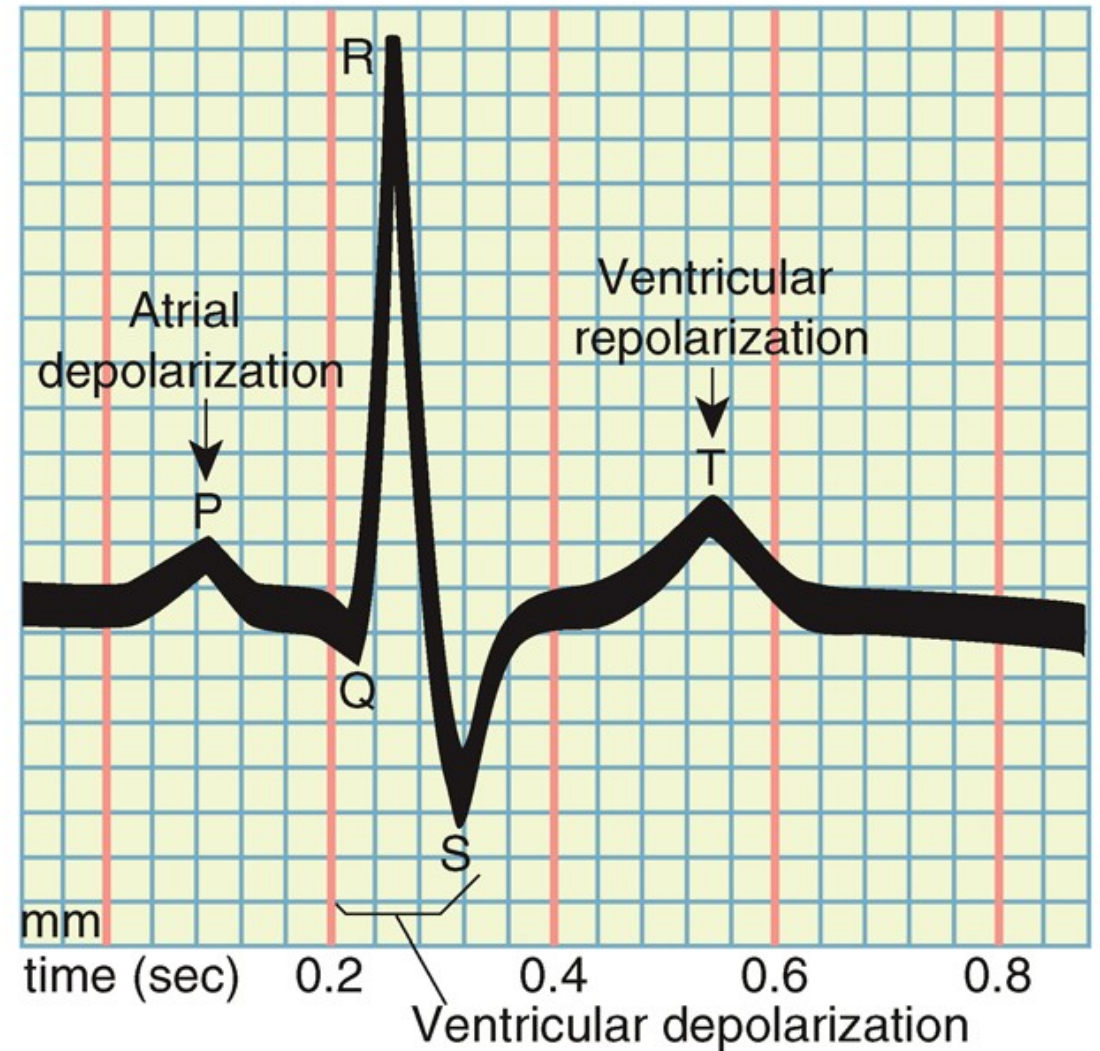
3. **Bundle of His:** located in upper interventricular septum
 - only electrical connection b/n atria & ventricles
4. **Bundle branches:** located in interventricular septum
5. **Purkinje fibers:** located in ventricular walls
 - Spreads the AP out so that the ventricle contract

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ECG: Electrocardiogram

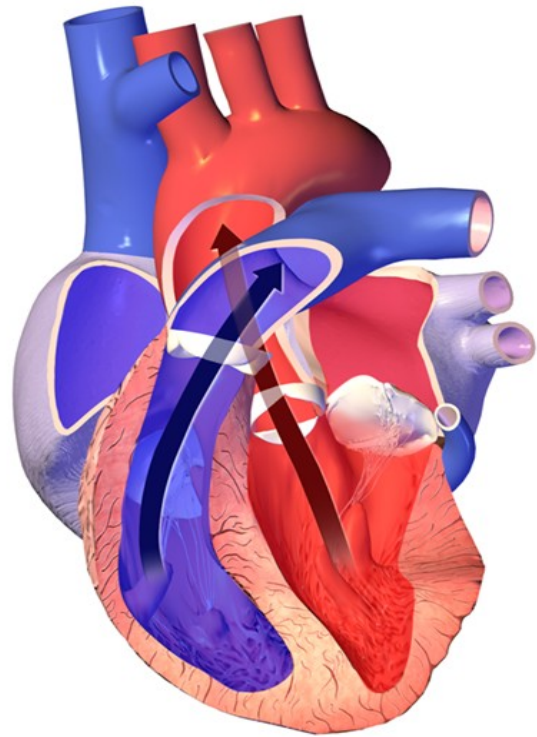
- P Wave: atrial depolarization
- QRS: ventricular depolarization
- T Wave: ventricular repolarization
- Depolarization= contraction
- Repolarization= relaxation



Cardiac Cycle:

Sequence of events in 1 heartbeat

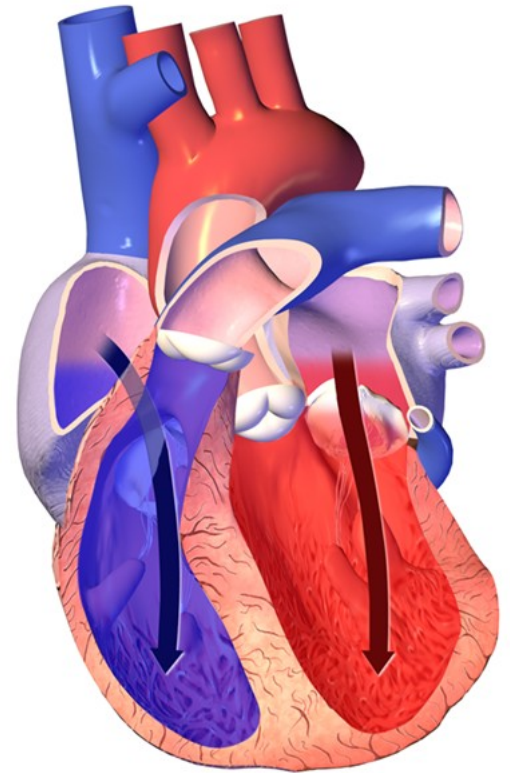
Simultaneous contraction of atria followed by ventricles followed by a rest period



Systole
(pumping)

SYSTOLE: CONTRACTION

DIASTOLE: RELAXATION



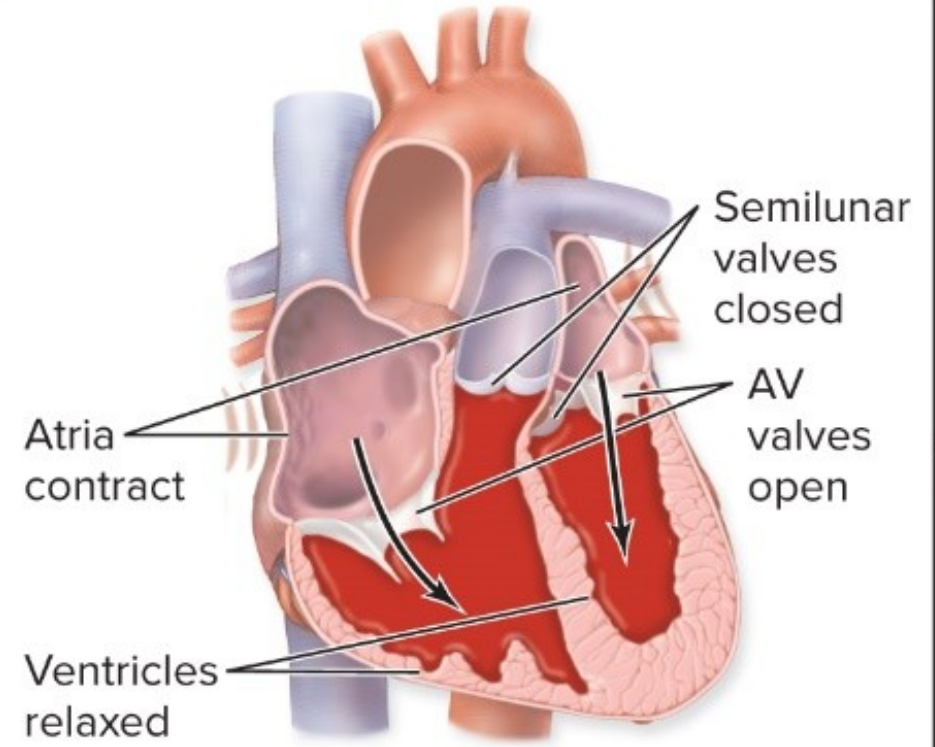
Diastole
(filling)

Atrial Systole

Atrial Systole: Pressure builds as more blood enters atriums causing them to contract

- “Tops Off” the ventricles: filling them to capacity (end-diastolic vol 130mL)
- Atrial diastole then occurs

1 Atrial contraction and ventricular filling

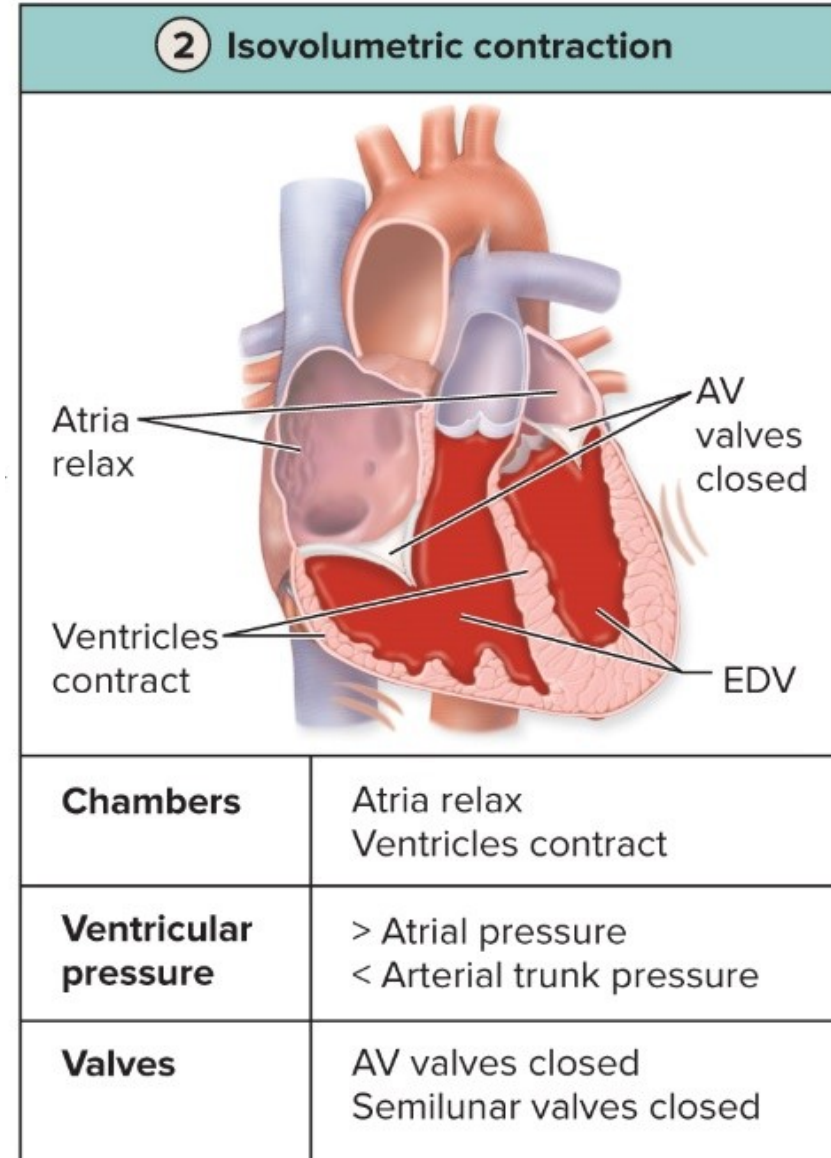


Chambers	Atria contract Ventricles relax
Ventricular pressure	< Atrial pressure < Arterial trunk pressure
Valves	AV valves open Semilunar valves closed

Ventricular Systole: Part 1

Initial pressure of ventricular systole (1st phase) forces Left & Right AV valves closed (but not enough pressure to eject blood into arteries)

Pressure continues to rise in ventricles until it exceeds pressure in arterial trunks



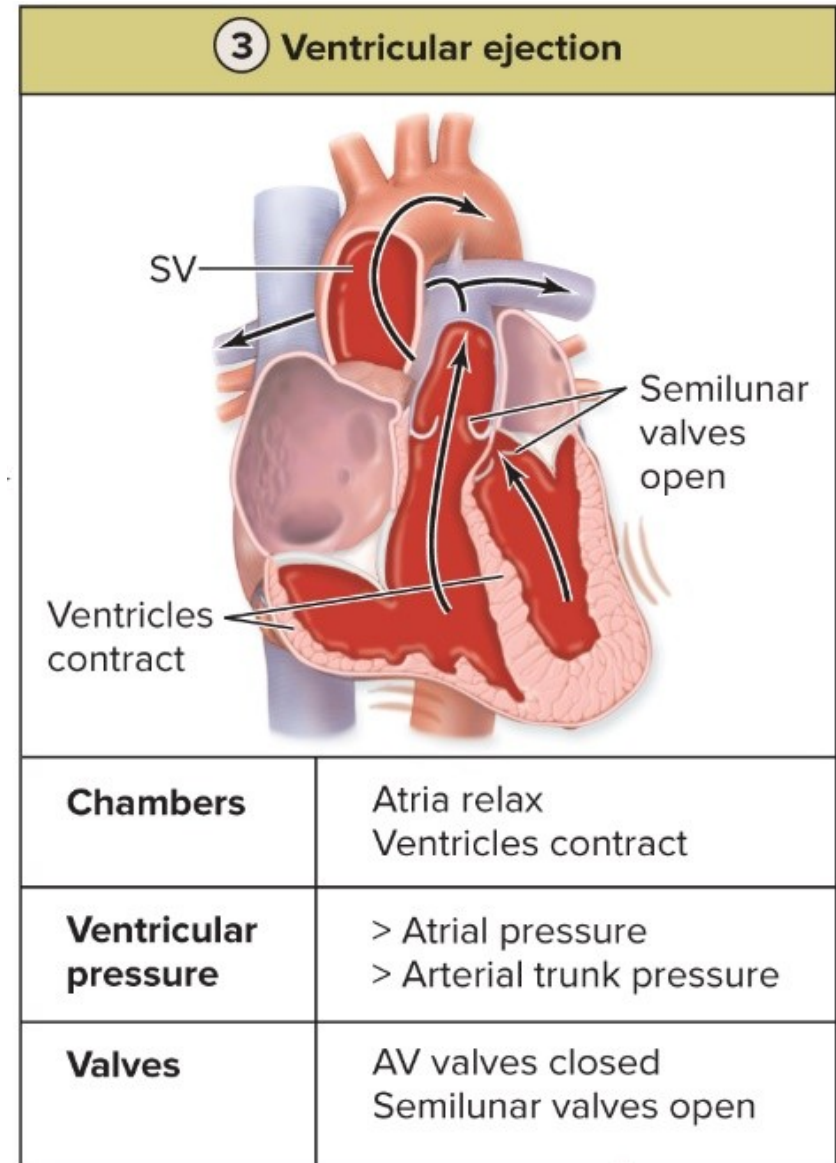
Ventricular Systole: Part 2

Ventricles enter second phase of systole & contract enough to send blood into aortic & pulmonary trunks (**ventricular ejection**)

Stroke Volume: each ventricle ejects 60-80mL of blood

Ejection fraction: % of blood in ventricle pumped out during ventricular systole (appx 60% of blood in ventricle)

End Systolic Volume (ESV): Amount of blood remaining in ventricle when semilunar valves close



*Ventricular Systole

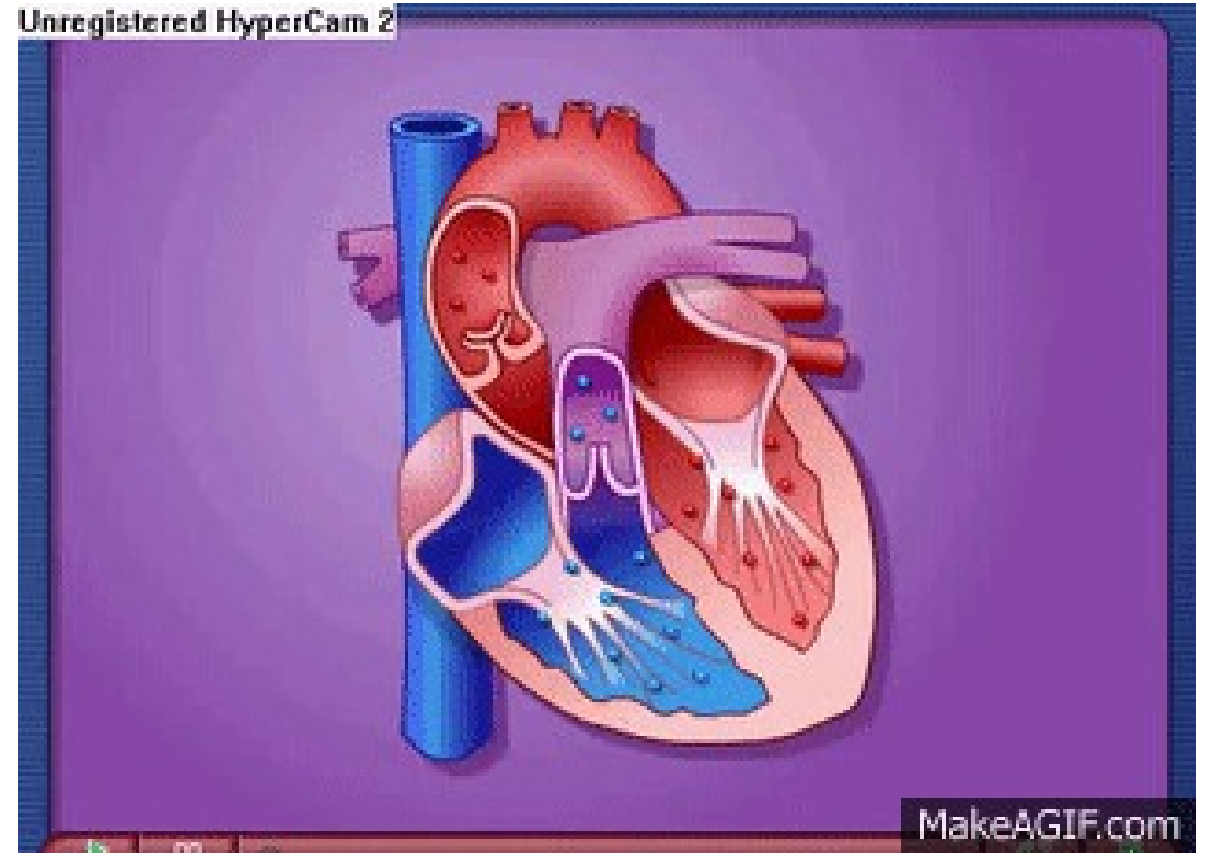
Ventricular Systole: contraction of ventricles

Ventricles enter systole at same time the atria enter diastole

What valves prevent backflow of blood into the atrium during contraction?

Left atrium= mitral valve

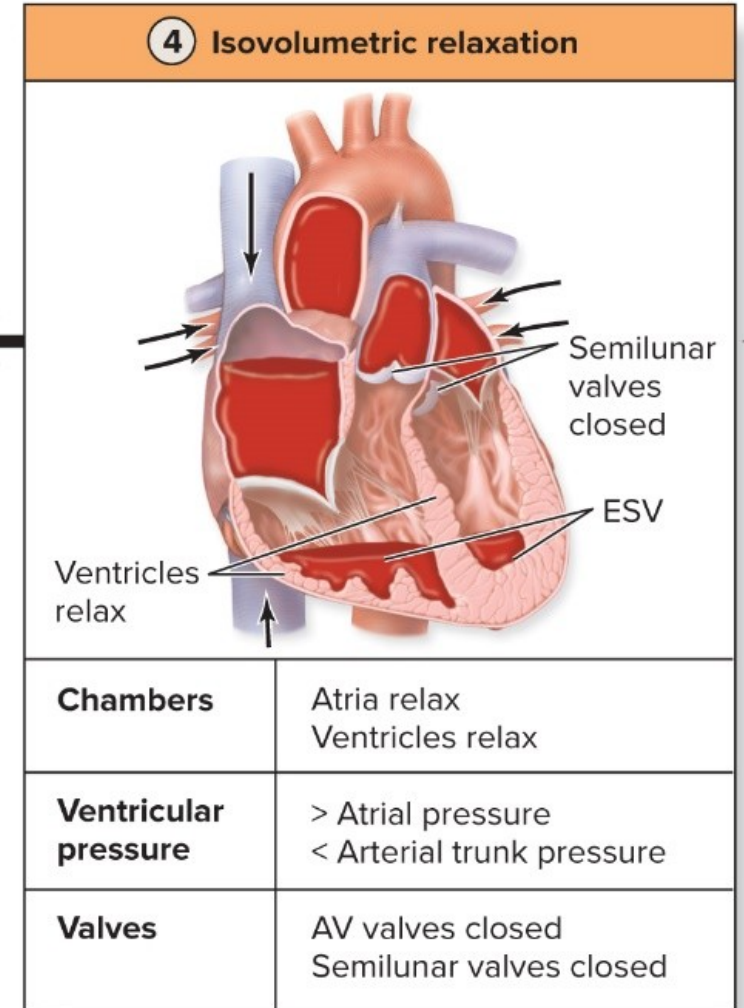
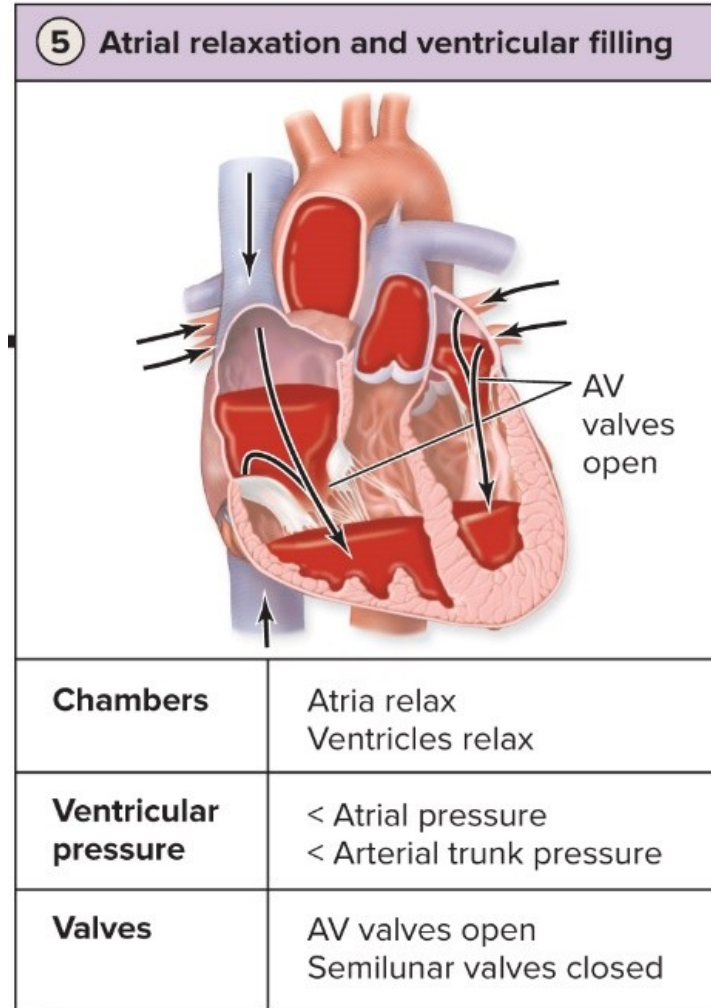
Right atrium= tricuspid valve



Ventricular Diastole

Atrial & Ventricular Diastole overlap to give heart a complete rest between beats

Blood is flowing passively into Atria & Ventricles during **DIASTOLE**



Cardiac Output

$$\text{HR (heart rate)} \times \text{SV (stroke volume)}$$

HR:

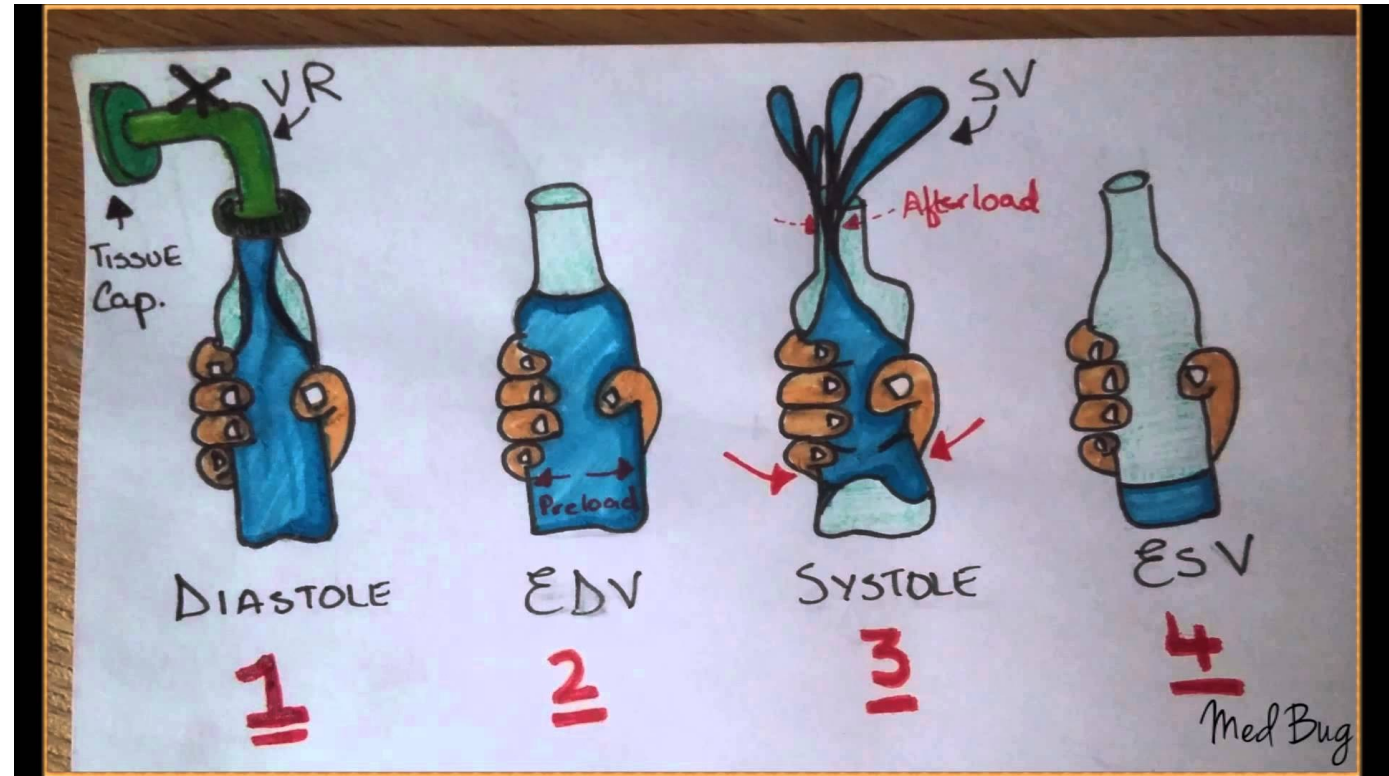
- Number of heart beats per minute

Stroke Volume:

- Amount of blood pumped out by a ventricle with each beat

Cardiac output (ml/min)

- Amount of blood pumped from the heart in one minute



What influences cardiac output?

Heart rate

- Increases HR and CO
 - Sympathetic nervous system
 - Exercise
 - Low BP
 - Low Oxygen
- Decreases HR and CO
 - Parasympathetic NS

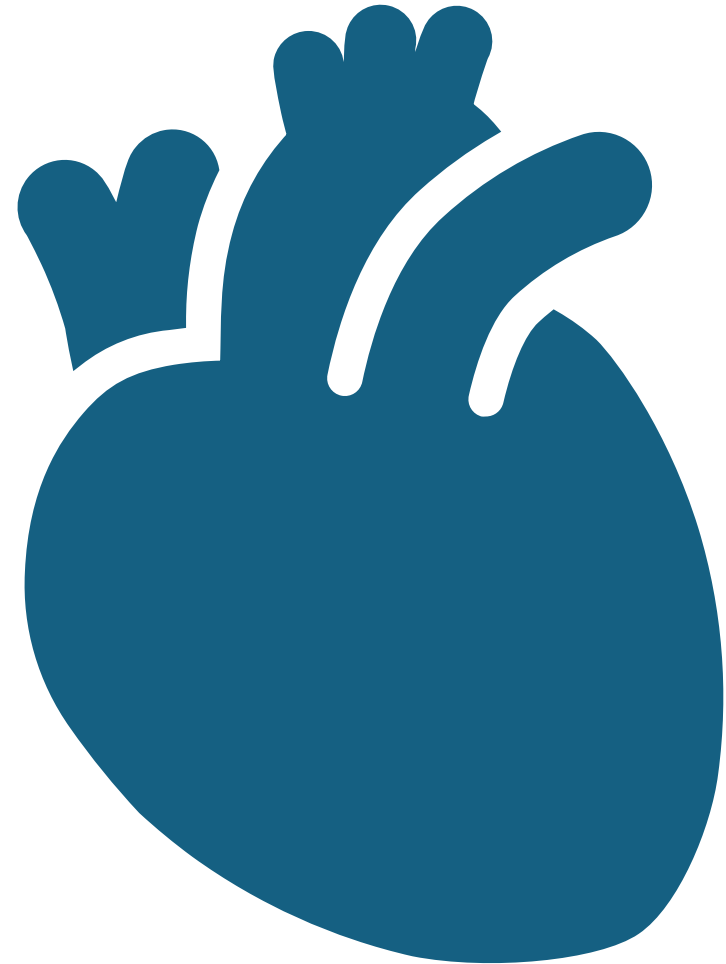
Stroke volume

- Increases SV and CO
 - Increased preload
 - Decreased afterload
 - Increase venous return
- Decreases SV and CO
 - Decreased preload- not enough blood i.e. dehydration, hemorrhage
 - Increased after load i.e. high blood pressure

Practice Question

S1 and S2 in the cardiac conduction cycle are directly due to which?

- A - ventricle contraction**
- B - valves closing**
- C - atria relaxation**
- D - blood rushing out of the heart**



EXAM PREP:

- Cardiac cycle: phases and what happens in each one.
- Function/job of valves and when they are closed. Which heart sound goes with which valves.
- Flow of blood through the heart – where does each part pump blood to?
- Cardiac output – how to calculate, what impacts this.
- Conduction system: parts and job – SA, internodal, AV, bundle of His, bundle branches, Purkinje fibers
- ECG: what does each wave represent?

